

TOP 15 PYTHON IN EXCEL FOR FINANCE TIPS



by Christian Martinez



Data Preparation & Cleaning

1) Adding your data to a dataframe

`df = xl("B4:C9", headers=True)`

Pro-tip: Set headers=True so the first row becomes your column names automatically.

Use the xl() function to pull data directly from your worksheet range into Python.

2) Outlier Detection & Removal

`df = df[df['Revenue_Amount'] < df['Revenue_Amount'].quantile(0.95)]`

Spot and exclude extreme values.

3) Missing Values

Fill gaps using median, mode, or advanced methods.

`df.fillna(df.median(numeric_only=True))`

Forecasting and Insights

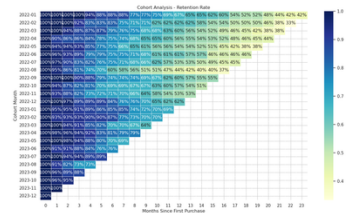
11) Revenue Forecasting

Assess forecast accuracy and model performance.



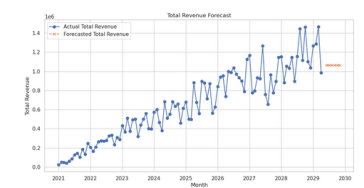
12) Cohort Retention Heatmap

Track how customer retention changes over time.



13) Machine Learning

Project future revenue using trend and seasonality models.



Descriptive Analytics & Diagnostics

4) Correlation Matrix for KPIs

`df_forecast.corr(numeric_only=True)`

Identify relationships between financial variables.

	Scenario #	Forecast Year	Revenue Assumption	Cost Assumption	Probability Weight	Net Profit Forecast
Scenario #	1.000	0.940	0.078	0.054	0.052	0.122
Forecast Year	0.940	1.000	0.062	0.063	0.034	0.048
Revenue Assumption	0.078	0.062	1.000	0.979	-0.488	0.873
Cost Assumption	0.054	0.063	0.979	1.000	-0.460	0.755
Probability Weight	0.052	0.034	-0.488	-0.460	1.000	-0.469
Net Profit Forecast	0.122	0.048	0.873	0.755	-0.469	1.000

6) Descriptive Statistics Summary

Summarize key metrics like mean, median, and standard deviation.

One full table of statistics with just one line!

	Scenario #	Forecast Year	Revenue Assumption	Cost Assumption
count	100	100	100	100
mean	50.5	2026.65	1301200	832200
std	29.01149198	0.845367651	316391.5232	235211.4029
min	1	2025	800000	500000
25%	25.75	2025	1037500	650000
50%	50.5	2026	1275000	800000
75%	75.25	2027	1500000	950000
max	100	2027	2100000	1400000

5) Variance & Trend Analysis

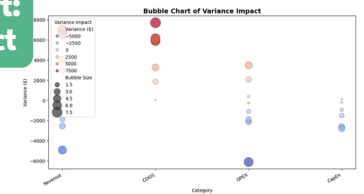
`budget_analysis['Variance'] = budget_analysis['Actual'] - budget_analysis['Budget']`

Measure variability and spot trends over time.

Category	Actual	Budget	Variance
Revenue	105000	100000	5000
COGS	45000	42000	3000

14) Bubble Chart: Variance Impact

Visualize the size and direction of variances by category. Big bubbles = bigger financial impact.



15) Word Cloud: Financial Commentary

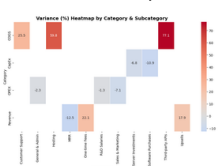
Quickly spot recurring themes in FP&A Commentary.



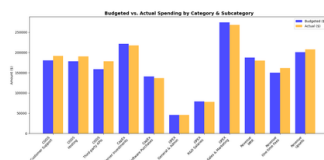
Analytics & Visualization

7) Variance Heatmap

Spot over/underperformance fast with a color-coded variance heatmap.

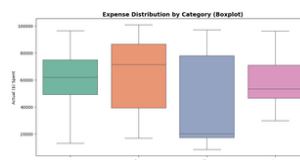


8) Bar Chart



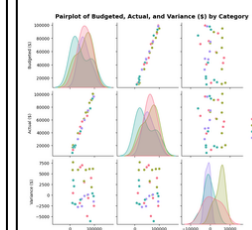
Compare budgeted vs. actual spend side by side.

9) Expense Distribution Boxplot



See spending spread and outliers across categories.

10) Analysis Pairplot



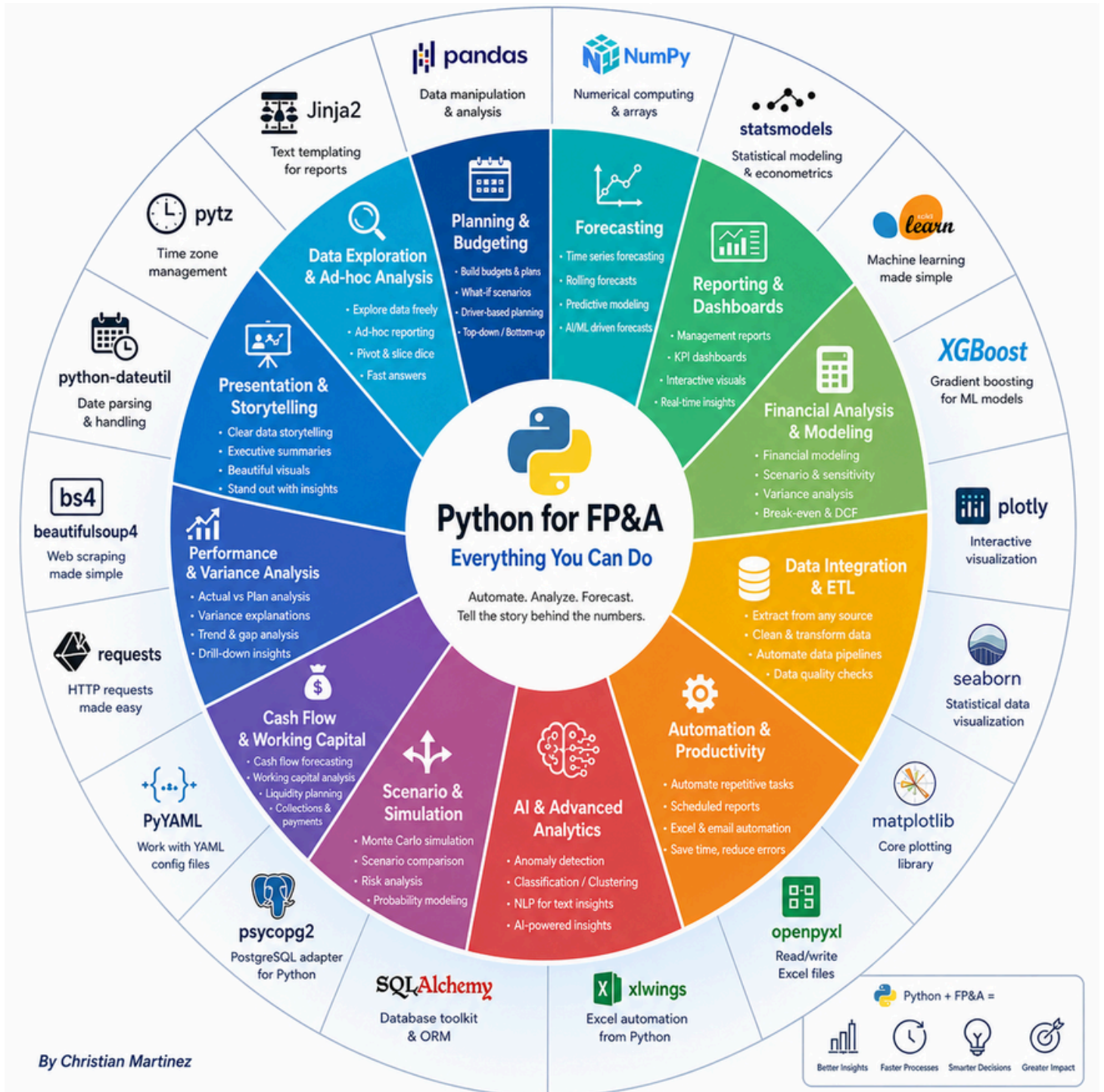
Explore trends across key metrics.

Send me a message to get all code!



PYTHON LIBRARIES GUIDE

by Christian Martinez



PYTHON LIBRARIES GUIDE

by Christian Martinez

#1

Pandas

The backbone of FP&A

Data cleaning, transformation, financial modeling, and time-series analysis. Essential for almost everything.

#2

NumPy

Fast numerical engine

Handles large-scale calculations, arrays, and financial computations efficiently.

#3

Matplotlib

Classic visualization tool

Create financial charts, trends, and budget vs. actual visuals.

#4

Seaborn

Beautiful statistical visuals

Great for dashboards, variance analysis, and executive-ready charts.

#5

OpenPyXL

Excel integration powerhouse

Read/write Excel files, automate reports, and format financial models.

#6

Plotly

Interactive dashboards

Build dynamic FP&A dashboards with drill-down capabilities.

#7

XlsxWriter

Advanced Excel reporting

Create polished financial reports with charts, formatting, and formulas.

#8

Scikit-learn

Forecasting & predictive modeling

Used for revenue forecasting, cost prediction, and scenario planning.

#9

Statsmodels

Statistical & econometric analysis

Time-series forecasting (ARIMA), regression models for financial insights.

#10

Streamlit

FP&A dashboards made easy

Turn financial models into interactive web apps for stakeholders.

✦ BONUS MENTIONS

PySpark

Big data financial analysis



Dash

Enterprise-level dashboards



Prophet

Easy time-series forecasting



FP&A JOURNEY



HOW PYTHON POWERS FP&A

by Christian Martinez



HOW PYTHON POWERS FP&A

KEY USE CASES

1. DATA AUTOMATION AND INTEGRATION



Automate data collection, cleaning, and integration from multiple sources.

- Import data from Excel, CSV, SQL databases, APIs, and cloud platforms
- Automate ETL processes
- Data validation and cleansing
- Handle large datasets efficiently
- Schedule recurring data refreshes
- Reduce manual errors and save time

KEY LIBRARIES

pandas, SQLAlchemy, openpyxl, requests

2. FINANCIAL MODELING AND SCENARIO ANALYSIS



Build flexible models and run what-if scenarios with ease.

- Driver-based financial models
- Scenario & sensitivity analysis
- Monte Carlo simulations
- Goal seek and optimization
- Dynamic assumptions and forecasts
- Version control for models
- Stress testing and downside analysis

KEY LIBRARIES

pandas, numpy, scipy, statsmodels, pulp

3. FORECASTING AND PREDICTION



Leverage data and machine learning to improve forecast accuracy.

- Time series forecasting
- Revenue, expense, cash flow forecasting
- Predictive modeling with ML
- Customer or product level forecasting
- Forecast accuracy tracking
- Anomaly detection
- Predictive scenario planning

KEY LIBRARIES

pandas, statsmodels, scikit-learn, prophet

4. BUDGETING AND PLANNING



Streamline budgeting and planning workflows across the organization.

- Budget templates and drivers
- Top-down & bottom-up budgeting
- Variance analysis (Budget vs Actual)
- Rolling forecasts
- Zero-based budgeting support
- What-if modeling for plans
- Collaboration and audit trails

KEY LIBRARIES

pandas, openpyxl, numpy, xlwings

5. REPORTING AND VISUALIZATION



Create insightful reports and dashboards that drive decisions.

- Automated financial reporting
- KPIs, metrics and trend analysis
- Management dashboards
- Interactive visualizations
- Commentary and insights automation
- Export to Excel, PDF, PowerPoint
- Real-time reporting from live data

KEY LIBRARIES

pandas, matplotlib, seaborn, plotly, dash

6. ANALYTICS AND INSIGHTS



Uncover insights and drive better business decisions.

- Profitability and margin analysis
- Customer & product profitability
- Cost analysis and benchmarking
- Cohort and segmentation analysis
- Working capital optimization
- Cash flow insights
- Root cause analysis

KEY LIBRARIES

pandas, numpy, scipy, scikit-learn

7. PROCESS AUTOMATION AND PRODUCTIVITY



Automate repetitive FP&A tasks and enhance team productivity.

- Automate recurring reports
- Email alerts and notifications
- File management and archiving
- Workflow orchestration
- Automated PowerPoint generation
- Integration with tools & APIs

- Use Python as an FP&A command center

KEY LIBRARIES

openpyxl, smtplib, schedule, jinja2, win32com, requests

BENEFITS



Save time



Improve accuracy



Enhance insights



Increase scalability



Empower FP&A teams

HOW PYTHON POWERS FP&A

by Christian Martinez

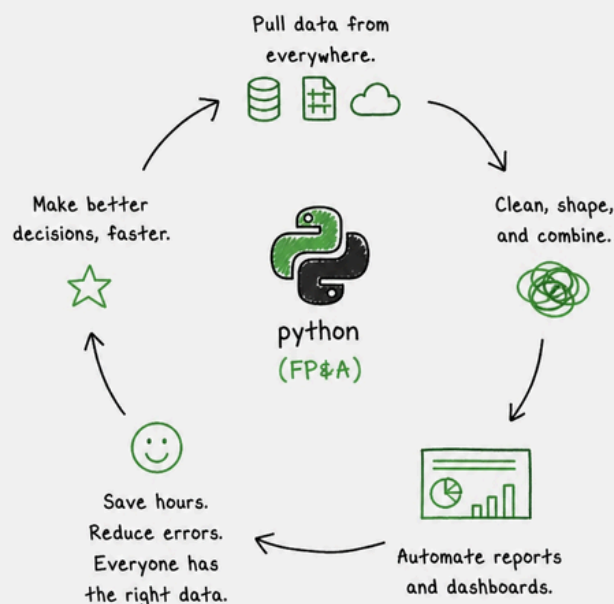
The FP&A Python Ecosystem

A CONNECTED PATH FROM DATA TO DECISIONS



Python for FP&A

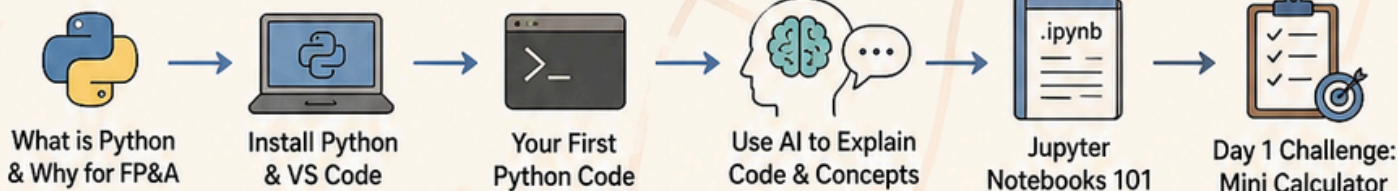
Automate the busywork.
Focus on the decisions.



PYTHON LEARNING ROADMAP

by Christian Martinez

Day 1 – Foundations & Setup



Day 2 – Data Skills for FP&A



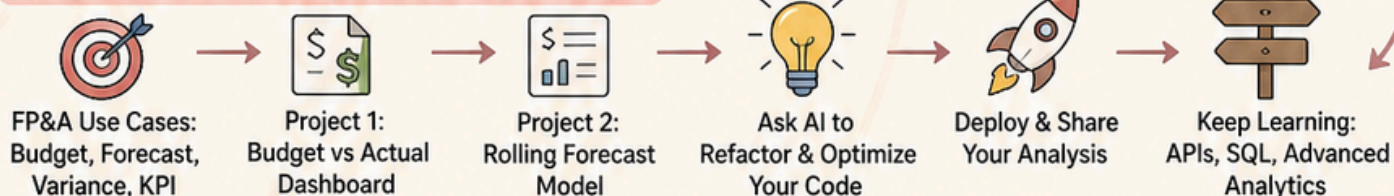
Day 3 – Analysis & Visualization



Day 4 – Automate FP&A Workflows



Day 5 – FP&A Projects & Next Steps



AI Prompts You'll Use Along the Way

- Explain this code step by step.
- Help me clean this dataset.
- Write Python code to calculate...
- Why is my result not correct?
- Improve this visualization.
- Make this code more efficient.

Success Looks Like:

- Automate repetitive tasks
- Deliver timely insights
- Tell the story behind the numbers
- Drive better business decisions



Google Colab for FP&A A Beginner's Guide

Google Colab is like Google Docs, but for Python.
Super easy to use, no install needed, and works in the cloud.
Perfect for FP&A professionals who want to analyze data,
build models, and create reports—anytime, anywhere.

by Christian Martinez

WHAT IS GOOGLE COLAB?

A free, cloud-based Jupyter notebook environment by Google that lets you write and run Python code in your browser—no installation required.



WHY FP&A PROFESSIONALS LOVE GOOGLE COLAB



No install needed



Works in the cloud



Easy to share and collaborate



Free access (with benefits)



Powerful Python environment



Secure and reliable

GET STARTED IN 6 EASY STEPS

- 1 Go to Google Colab**
Visit colab.research.google.com and sign in with your Google account.
- 2 Create a New Notebook**
Click "New Notebook" to open a fresh Colab notebook.
- 3 Rename Your Notebook**
Click the title ("Untitled.ipynb") to give it a meaningful name.
- 4 Write Code in Cells**
Type Python code in a cell and press Shift + Enter to run it.
- 5 Save to Google Drive**
Your notebook is automatically saved in your Drive.
- 6 Share and Collaborate**
Click "Share" to invite others to view or edit your notebook.

COLAB CORE FEATURES YOU'LL USE EVERY DAY

- Code Cells**
Write and run Python code.
- Markdown Cells**
Add notes, headers, and documentation.
- Run Cell**
Execute code with Shift + Enter or ▶ button.
- File Explorer**
Access and manage files in your environment.
- Upload Files**
Upload data from your computer or Drive.
- Connect to Data**
Easily read from Excel, CSV, Google Sheets, and more.

	Month	Revenue	Expense	Net Income
1	Jan	1200000	800000	400000
2	Feb	1150000	750000	400000
3	Mar	1400000	900000	500000
4	Apr	1500000	950000	550000
5	May	1600000	970000	630000



TIP: Use Markdown cells to explain your analysis and make your notebook clear, professional, and easy to follow.

POPULAR FP&A USE CASES IN COLAB



Budget vs Actual Analysis
Compare performance and calculate variances.



Forecasting
Build time series forecasts with Prophet or statsmodels.



Scenario Analysis
Model best, base, and worst case scenarios.



Dashboard Creation
Create interactive dashboards with Plotly or Panel.



Automated Reporting
Generate reports and export to Excel or Google Sheets.



Data Integration
Pull data from APIs, databases, or Google Sheets.

WORKING WITH DATA IN COLAB (EXAMPLES)

```
# Read Excel file from Google Drive
df = pd.read_excel('/content/drive/MyDrive/FP&A/Actuals.xlsx')

# Read CSV file
df = pd.read_csv('/content/FP&A/Forecast.csv')

# Connect to Google Sheets
import gspread
from google.colab.auth import authenticate_user
from google.colab import default

# Quick summary
df.describe()

# Create a chart
df.groupby('Month')['Revenue'].sum().plot(kind='line', title='Revenue Trend')
```

COLAB SHORTCUTS CHEAT SHEET

Shift + Enter	Run cell
Ctrl + Enter	Run cell
Alt + Enter	Run cell & insert below
Ctrl + /	Comment / uncomment
Esc then A	Insert code cell above
Esc then B	Insert code cell below
Esc then M	Insert Markdown cell
Ctrl + S	Save notebook
Ctrl + F	Find and replace

BEST PRACTICES

- ✓ Keep each cell focused on one task.
- ✓ Use Markdown to document your steps.
- ✓ Name your files and variables clearly.
- ✓ Save versions and collaborate responsibly.
- ✓ Use Drive folders to keep things organized.
- ✓ Monitor memory usage with large datasets.



TIP: Free users get access to CPUs. Upgrade to Colab Pro for more RAM, faster GPUs, and longer runtimes.

COLAB PRO VS FREE (AT A GLANCE)

Feature	Free	Pro
Price	\$0	From ~\$9.99 / month
RAM	Up to ~12 GB	Up to ~25 GB+
GPUs	Limited access	Priority access
Runtime	Shorter sessions	Longer sessions
Best for	Learning, small projects	Larger analysis & models

USEFUL RESOURCES

- Colab Help Center**
https://colab.research.google.com/notebooks/basic_features_overview.ipynb
- Colab GitHub**
<https://github.com/googlecolab/colabtools>
- Python for Data Analysis**
<https://wesmckinney.com/book/>
- Official Docs**
<https://pandas.pydata.org/docs/>

READY TO START?

Open Google Colab and create your first notebook now:

colab.research.google.com



No installation. No hassle.
Just insights.

"The more you practice in Colab, the more value
you'll unlock for your business."

— CM —



TOOLS TODAY. INSIGHT TOMORROW. IMPACT ALWAYS.



How to use Python for FP&A 101 Start here.

*A beginner-friendly roadmap to start using Python
and add immediate value in FP&A.*

by Christian Martinez

Python is a powerful, flexible, and easy-to-learn tool that helps FP&A professionals automate tasks, analyze data, build forecasts, and create impactful dashboards.



Save time and
automate
repetitive work



Analyze data
faster and
deeper



Improve forecast
accuracy and
scenario planning



Connect and
work with any
data source



Build dashboards
and reports
that stand out

THE FP&A PYTHON STARTER ROADMAP

- 1 Learn the Basics**
Get comfortable with Python fundamentals.
Focus on: variables, data types, lists, dictionaries, loops, functions, and conditionals.
- 2 Work with Data (pandas)**
Import, clean, and transform your financial data.
Learn to read Excel/CSV, filter, group, aggregate, pivot, and merge data.
- 3 Analyze & Visualize**
Turn data into insights with visuals.
Use matplotlib and plotly to create charts and dashboards that tell the story.
- 4 Build Forecasts & Models**
Use Python to build financial models and forecasts.
Leverage libraries like numpy, scipy, statsmodels, prophet, and sklearn.
- 5 Automate & Integrate**
Automate reports, connect APIs, and schedule tasks.
Use Python to pull data, run processes, and send reports automatically.
- 6 Build Dashboards & Reports**
Create interactive dashboards and share insights.
Use Streamlit, Plotly Dash, or export to Excel/HTML for easy sharing.

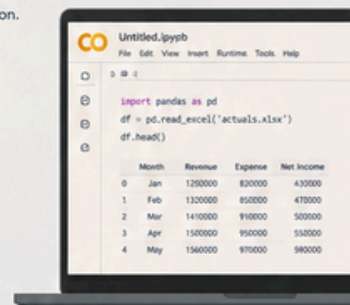
OUR #1 RECOMMENDATION: USE GOOGLE COLAB



Google Colab is like Google Docs but for programming in Python.

- ✓ Super easy to use
- ✓ No install needed
- ✓ Free and works in the cloud
- ✓ Access your work anywhere
- ✓ Share with your team instantly

Perfect for FP&A professionals who want to start fast and stay productive.



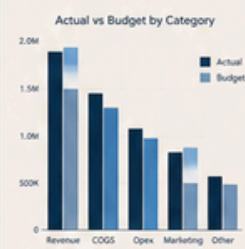
EXAMPLE: READ, ANALYZE & VISUALIZE IN MINUTES

```
import pandas as pd
import plotly.express as px

# 1. Load data
df = pd.read_excel('budget_vs_actual.xlsx')

# 2. Analyze
summary = df.groupby('Category')[['Actual', 'Budget']].sum()
summary['Variance'] = summary['Actual'] - summary['Budget']

# 3. Visualize
fig = px.bar(summary, x=summary.index, y=['Actual', 'Budget'],
             barmode='group', title='Actual vs Budget by Category')
fig.update_layout(template='plotly_white')
fig.show()
```



ESSENTIAL PYTHON LIBRARIES FOR FP&A



pandas

Data manipulation
and analysis



numpy

Numerical operations
and arrays



matplotlib

Static charts and
visualizations



plotly

Interactive charts
and dashboards



openpyxl

Read/write Excel
files



requests

Call APIs and get
external data



prophet

Time series forecasting
made easy

BEST PRACTICES

- ✓ Keep your data clean and consistent
- ✓ Comment your code and use clear names
- ✓ Break tasks into small, reusable functions
- ✓ Validate results and check for outliers
- ✓ Automate repetitive tasks and save time
- ✓ Document your processes for your team



TIP

Start small, stay consistent,
and build as you go.
The more you practice,
the more value you'll
unlock for your business.

WHAT YOU CAN DO FIRST

- Pull actuals and budget data from Excel
- Build a variance analysis in pandas
- Create a simple chart to visualize results
- Automate a monthly report
- Share your analysis with your team

*"You don't need to be a developer—
you just need to be curious and start."*

— CM —



TOOLS TODAY. INSIGHT TOMORROW. IMPACT ALWAYS.



7 Best Ways to Use Matplotlib Library for Visualizing FP&A Data

Turn complex financial data into clear, actionable insights.

by Christian Martinez

Matplotlib gives FP&A professionals the power to build clear, custom, and publication-ready visualizations—exactly the way you need them.



Highly customizable and flexible



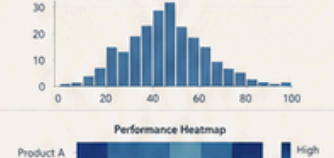
Integrates seamlessly with Python stack



Great for static, high-quality charts



Perfect for reporting, presentations, and dashboards

7 BEST WAYS	WHAT IT SHOWS	HOW TO DO IT (PYTHON)	EXAMPLE
01  Line Chart for Trends Over Time	Track KPIs, revenue, expenses, or cash flow trends over time.	<pre>plt.plot(df['Date'], df['Revenue']) plt.title('Revenue Over Time') plt.xlabel('Date') plt.ylabel('Revenue (USD)') plt.grid(True)</pre>	
02  Bar Chart for Category Comparison	Compare actual vs budget, or performance across business units.	<pre>plt.bar(x, actual, label='Actual') plt.bar(x, budget, label='Budget', alpha=0.7) plt.legend() plt.title('Actual vs Budget')</pre>	
03  Stacked Bar Chart for Composition Analysis	Visualize how each component contributes to the total.	<pre>plt.bar(x, df['COGS'], label='COGS') plt.bar(x, df['Opex'], bottom=df['COGS'], label='Opex') plt.bar(x, df['Other'], bottom=df['COGS']+ df['Opex'], label='Other') plt.legend(); plt.title('Expense Breakdown')</pre>	
04  Pie Chart for Part-to-Whole View	Show percentage contribution of categories to the total.	<pre>plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90) plt.title('Revenue by Segment')</pre>	
05  Scatter Plot for Relationship Analysis	Explore relationships between drivers—e.g., price vs volume vs revenue.	<pre>plt.scatter(df['Price'], df['Volume'], c=df['Revenue'], cmap='Blues') plt.xlabel('Price (USD)') plt.ylabel('Volume') plt.colorbar(label='Revenue') plt.title('Price vs Volume (Colored by Revenue)')</pre>	
06  Histogram for Distribution Analysis	Understand the distribution of values—e.g., deal sizes, days sales outstanding.	<pre>plt.hist(df['DSO'], bins=20, color='4C78A8', edgecolor='white') plt.xlabel('Days Sales Outstanding') plt.ylabel('Frequency') plt.title('DSO Distribution')</pre>	
07  Heatmap for Pattern Detection	Identify patterns in performance across time and categories.	<pre>im = plt.imshow(df.values, cmap='YlGnBu', aspect='auto') plt.colorbar(label='Value') plt.xticks(range(len(df.columns)), df.columns, rotation=45, ha='right') plt.yticks(range(len(df.index)), df.index) plt.title('Performance Heatmap')</pre>	

BEST PRACTICES

- ✓ Keep it simple and focused.
- ✓ Use clear titles, labels, and units.
- ✓ Highlight key insights, avoid clutter.
- ✓ Choose the right chart for the story.
- ✓ Consistency in colors and formatting.



TIP
Matplotlib is the foundation of many advanced visualizations in Python. Master the basics, then customize to tell your FP&A story your way.

WHEN TO REACH FOR MATPLOTLIB

- 📊 You need full control over your visuals
- 📄 You're creating static reports or presentations
- 🔗 You want lightweight and dependency-friendly charts
- 🛠️ You need to customize beyond default charting tools
- 🤖 You're building automated reports in Python

"Good visualization doesn't just show numbers—it helps finance leaders see the story behind them."

— CM —



TOOLS TODAY. INSIGHT TOMORROW. IMPACT ALWAYS.



How to use Prophet Library for Forecasting FP&A Data



Accurate forecasts. Smarter decisions.

by Christian Martinez

Prophet is a forecasting library developed by Meta. It's built for business time series—handling seasonality, holidays, and trends with minimal effort and maximum interpretability.



Handles seasonality, holidays, and trends automatically



Robust to missing data and structural breaks



Highly interpretable and easy to customize



Fast to implement and production ready

STEP-BY-STEP GUIDE

1 Install Prophet

```
pip install prophet
```

2 Prepare Your Data

Prophet expects a dataframe with two columns: **ds** (date) and **y** (value).

```
import pandas as pd
df = pd.DataFrame({
    'ds': pd.date_range('2021-01-01', periods=100, freq='D'),
    'y': your_values
})
```

3 Create and Fit the Model

```
from prophet import Prophet
m = Prophet()
m.fit(df)
```

4 Make a Forecast

```
future = m.make_future_dataframe(periods=90) # 90 days ahead
forecast = m.predict(future)
```

5 Visualize the Results

```
from prophet.plot import plot_plotly, plot_components_plotly
fig1 = plot_plotly(m, forecast)
fig2 = plot_components_plotly(m, forecast)
fig1.show()
fig2.show()
```

6 Analyze and Use the Forecast

Use `forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']]` to integrate into reports, dashboards, and scenario models.

EXAMPLE FORECAST



Prophet captures trend and seasonality, then projects forward with confidence intervals.

KEY CAPABILITIES



Trend Detection

Automatically models changing trends over time.



Seasonality

Captures daily, weekly, monthly, and yearly seasonality.



Holidays & Events

Incorporate holidays and business events that impact performance.



Custom Regressors

Add drivers like prices, marketing spend, or FX rates to improve accuracy.



Uncertainty Intervals

Quantify forecast confidence with upper and lower bounds.



Missing Data Friendly

Handles missing values and irregular time intervals gracefully.

BEST PRACTICES

- ✓ Start simple. Use default settings, then add complexity as needed.
- ✓ Validate your forecasts with backtesting.
- ✓ Include holidays and known events.
- ✓ Use relevant regressors to improve accuracy.
- ✓ Regularly retrain your model with new data.



TIP

Prophet works best with clean, consistent data and a clear understanding of your business drivers.

COMMON FP&A USE CASES

- Revenue forecasting and rolling forecasts
- Budgeting and reforecasting cycles
- Cash flow and liquidity forecasting
- Demand planning and scenario analysis
- Financial KPI and operational forecasting

Why FP&A teams love Prophet

It's powerful, flexible, and built for business realities—giving you more time to analyze and less time to wrangle data.

“The best forecasts don't just predict the future—they prepare your business for it.”

— CM —



TOOLS TODAY. INSIGHT TOMORROW. IMPACT ALWAYS.



The 4 Levels of Data Analytics for FP&A

How to use Python at each level to drive insight, impact, and value.

by Christian Martinez



Analytics maturity in FP&A isn't about tools—it's about the questions you can answer. Progress through these 4 levels to transform data into a strategic advantage.

LEVEL	FOCUS	HOW PYTHON HELPS	KEY LIBRARIES	EXAMPLE USE IN FP&A
01 DESCRIPTIVE What happened? 	Understand historical performance through reporting and basic summaries.	Python automates data extraction, cleaning, and aggregation—saving time and improving report accuracy.	 pandas  openpyxl  matplotlib / seaborn	Automate budget vs. actual reports, variance analysis, and KPI dashboards.
02 DIAGNOSTIC Why did it happen? 	Identify drivers and root causes behind performance variances.	Python enables deeper exploration of data to uncover patterns, correlations, and key business drivers.	 pandas  numpy  seaborn / plotly  scipy.stats	Analyze price vs. volume impact, mix shifts, and cost variances to explain performance.
03 PREDICTIVE What will happen? 	Forecast future outcomes using statistical models and machine learning.	Python provides the modeling power to forecast with greater accuracy and adapt to changing conditions.	 statsmodels  scikit-learn  prophet  numpy	Build revenue forecasts, cash flow projections, and scenario models with confidence bands.
04 PRESCRIPTIVE What should we do? 	Recommend actions and optimize decisions to achieve business goals.	Python enables optimization, simulation, and what-if analysis to recommend the best course of action.	 scipy.optimize  pulp / pyomo  numpy  plotly	Optimize pricing, allocate budgets, and run scenario simulations to maximize financial outcomes.

HOW TO PROGRESS



Start with clean, trusted data



Answer the right questions



Automate and model with Python



Turn insights into decisions and impact



From reporting to recommendation—each level builds on the last. Python is the bridge that connects them all.

*“Data tells you what happened.
Analytics tells you what to do next.”*

— CM —



TOOLS TODAY. INSIGHT TOMORROW. IMPACT ALWAYS.



THE TOP 10

Python Libraries for FP&A



by Christian Martinez

Modern tools for modern finance teams. Leverage Python to automate, analyze, and elevate your FP&A function.



Drive better decisions



Save time with automation



Integrate seamlessly



Scale with confidence

	LIBRARY	WHAT IT DOES	USE IN FP&A FOR
01	 pandas <code>import pandas as pd</code>	Data manipulation and analysis for structured data.	Financial reporting, variance analysis, consolidations
02	 NumPy <code>import numpy as np</code>	High-performance numerical computing and arrays.	Forecast modeling, simulations, large data calculations
03	 openpyxl <code>import openpyxl</code>	Read, write, and modify Excel files programmatically.	Automate reports, templates, and data imports/exports
04	 matplotlib <code>import matplotlib.pyplot as plt</code>	Create static, animated, and interactive visualizations.	Dashboards, charts, KPIs, trend analysis
05	 seaborn <code>import seaborn as sns</code>	Statistical data visualization built on matplotlib.	Distributions, correlations, heatmaps, insights
06	 statsmodels <code>import statsmodels.api as sm</code>	Statistical modeling and econometric analysis.	Regression analysis, forecasting, driver-based planning
07	 SQLAlchemy <code>from sqlalchemy import create_engine</code>	Database toolkit and ORM for seamless data access.	Pull financial data, build pipelines, integrate systems
08	 plotly <code>import plotly.express as px</code>	Interactive, web-ready visualizations.	Executive dashboards, interactive analysis
09	 requests <code>import requests</code>	Send HTTP requests with ease.	Pull data from APIs, financial data feeds
10	 python-dateutil <code>from dateutil import parser</code>	Powerful extensions for datetime handling.	Time intelligence, forecasting calendars, period logic



Python is more than a coding language—it's a *force multiplier* for finance teams. Choose the right tools. Build with intent. Lead with insight.

"The best FP&A teams don't just report the past—they model the future."

— CM —



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